
BIOLOGICAL SAFETY EVALUATION PLAN

CLINISPONGE®/TIDESPON

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1. Purpose

To plan evaluation of the biocompatibility of the device Clinisponge®/Tidespon.

2. Scope

This plan covers the following device variants in its scope.

Table 1: Scope of Clinisponge®/Tidespon

Sr. No.	Device Model	Size
1.	CLINISPONGE®/TIDESPON Standard	80 x 50 x 10 mm 70 x 50 x 10 mm
2.	CLINISPONGE® /TIDESPON Special	80 x 50 x 1 mm 70 x 50 x 1 mm 125 x 80 x 1 mm 250 x 80 x 1 mm
3.	CLINISPONGE®/TIDESPON Film	200 x 70 x 0.5 mm 80 x 50 x 0.5 mm
4.	CLINISPONGE®/TIDESPON Dial	30 x 30 x 10 mm 40 x 10 x 10 mm
5.	CLINISPONGE®/TIDESPON Tampon	80 x Ø 30 mm 60 x Ø 20 mm
6.	CLINISPONGE® /TIDESPON Dental	10 x 10 x 10 mm 14 x 7 x 7 mm
7.	CLINISPONGE® /TIDESPON Long	125 x 80 x 10 mm
8.	CLINISPONGE® /TIDESPON Strip	80 x 20 x 10 mm
9.	CLINISPONGE® /TIDESPON Large	250 x 80 x 10 mm
10.	CLINISPONGE® /TIDESPON Nasal	50 x 10 x 10 mm
11.	CLINISPONGE® /TIDESPON Powder	1 gram 2 gram

3. Reference

- ISO 10993-1:2018 – Biological Evaluation of medical devices – Part-1: Evaluation and testing within a risk management process
- MDR 2017/745: Medical device regulation of the European parliament and of the council of 5 April 2017 on medical devices
- ISO 14971:2019 Medical Devices – Application of risk management to medical devices, Annex I: Guidance on Risk Analysis Procedures for Biological Hazards
- US-FDA Guidance Use of International Standard ISO 10993-1, "Biological evaluation of medical devices - Part 1: Evaluation and testing within a risk management process" of September 2020
- ISO 22442-1:2020 – Medical devices utilizing animal tissue and their derivatives Paer 1: Application of risk management

4. Definitions

Table 2: Common definitions used in the document

Biocompatibility	Ability of a medical device or material to perform with an appropriate host response in a specific application
Biological Risk	Combination of the probability of harm to health occurring because of adverse reactions associated with medical device or material interactions, and severity of that harm
Direct Contact	Medical device or medical component that comes into physical contact with body tissue
Externally Communicating Medical Devices	Medical device or medical device components that is partially or wholly located outside the body but has either direct or indirect contact with the internal body fluids and/or tissues
Indirect Contact	Medical device or medical device component through which a fluid or gas passes, prior to the fluid or gas coming into physical contact with body tissue
Final Product	Medical device or medical device component that has been subjected to all manufacturing processes for the “to be marketed” medical device including packaging and if applicable, sterilisation
Material Characterisation	Broad and general process of collecting existing information about a material’s chemistry, structure, and other properties, and if appropriate, new data, to facilitate the evaluation of these properties
High Risk	Likely to happen, often, frequently, always
Medium Risk	Can happen, but not frequently
Low Risk	Unlikely to happen, rare, remote

5. Device description

Gelatine-based absorbable hemostat is a medical device used for controlling bleeding during or after surgery. The material and its design promote blood clotting process and facilitate hemostasis at the site of application. Gelatine is a widely used material for hemostats and is clinically proven for its performance and safety. Gelatine has been certified as generally recognised as safe (GRAS) by the USFDA and is compliant with US, European and Japanese pharmacopoeia. Gelatine is produced from the skin, horns, and/or hoofs of bovine or porcine sources.

The CLINISPONGE®/TIDESPON is made from Type A Gelatin, formaldehyde and distilled water. CLINISPONGE®/TIDESPON is designed to for surgical procedures to stop capillary, venous bleeding and oozing where conventional methods are impractical and inefficient and use of non-absorbable materials are undesirable. The texture of the CLINISPONGE®/TIDESPON is smooth and flexible. Clinisponge®/Tidespon is absorbed completely within 3-4 weeks after implantation.

Table 3: Device information

Medical Device Name	CLINISPONGE®/TIDESPON	
Generic Name	Absorbable Haemostatic Sponge	
Device Class	Class III	
Device Classification Rule	Rule 18, Indent 0 of Annex VIII of the EU Medical Device Regulation 2017/745	
Basic UDI-DI	Product Variant	Basic UDI-DI
	CLINISPONGE®/TIDESPON Standard 80x50x10 mm	8698915380045JM
	CLINISPONGE®/TIDESPON Standard 70x50x10 mm	8698915380113JD
	CLINISPONGE®/TIDESPON Special 80x50x1 mm	8698915380069K3
	CLINISPONGE®/TIDESPON Special 70x50x1 mm	8698915380014JA
	CLINISPONGE®/TIDESPON Special 125x80x1 mm	8698915380CL1201HV
	CLINISPONGE®/TIDESPON Special 250x80x1 mm	8698915380CL2581KB

	CLINISPONGE®/TIDESPON Film 200x70x0.5 mm	8698915380182JY
	CLINISPONGE®/TIDESPON Film 80x50x0.5 mm	8698915380CL8055KT
	CLINISPONGE®/TIDESPON Dial 30x30x10 mm	8698915380199KH
	CLINISPONGE®/TIDESPON Dial 40x10x10 mm	8698915380CL4010J9
	CLINISPONGE®/TIDESPON Tampon 80xØ30 mm	8698915380052JJ
	CLINISPONGE®/TIDESPON Tampon 60xØ20 mm	8698915380CL6020J5
	CLINISPONGE®/TIDESPON Dental 10x10x10 mm	8698915380038JQ
	CLINISPONGE®/TIDESPON Dental 14x7x7 mm	8698915380007JD
	CLINISPONGE®/TIDESPON Long 125 x 80 x 10 mm	8698915380CL1210HW
	CLINISPONGE®/TIDESPON Strip 80 x 20 x 10 mm	8698915380CL8020K8
	CLINISPONGE®/TIDESPON Large 250 x 80 x 10 mm	8698915380CL2580K9
	CLINISPONGE®/TIDESPON Nasal 50 x 10 x 10 mm	8698915380CL5010JG
	CLINISPONGE®/TIDESPON Powder 1 gram	8698915380205JK
	CLINISPONGE®/TIDESPON Powder 2 gram	8698915380CL2000HQ
EMDN Code	M040503	
EMDN Description	GELATINE SPONGE HAEMOSTATIC DRESSINGS	

5.1 Intended use

Clinisponge®/Tidespon is a sterile local hemostatic sponge used for surgical procedures to stop capillary, venous bleeding and oozing where conventional methods are impractical and inefficient

and use of non-absorbable materials are undesirable. Clinisponge®/Tidespon may be used dry or soaked with sterile sodium chloride solution and with antibiotics, chemotherapeutic medications or thrombin without any reduction of the sponge's hemostatic effect.

Patient Target Group:

Patients who need to undergo one or more of the following surgeries: thorax surgery, parenchymal surgery, otorhinolaryngology, laminectomy, laparoscopy, cosmetic surgery, neurosurgery, vascular surgery, hemorrhoidectomies, anal fistula, fissures, dental/oral surgery, mandibular /maxillary surgeries.

Intended Users:

The product user profile is experienced and skilled medical practitioners for intended application.

5.2 Mechanism of action

Clinisponge®/Tidespon achieves hemostasis through a mechanical and physical mode of action. Its porous gelatine structure rapidly absorbs blood, concentrates platelets and coagulation factors, and accelerates the natural clotting process. Hemostasis is established by mechanical tamponade and support of thrombus formation. The sponge is subsequently absorbed by the body through natural enzymatic and phagocytic processes, eliminating the need for removal.

6. Biological Evaluation Strategy

6.1 Identification of materials and components

Gelatine, a gelling agent, formaldehyde, a homogenizer and cross-linking agent make up Clinisponge®/Tidespon.

Gelatine is an excellent component of hemostats because of its biodegradability and biocompatibility, high availability of raw material, low cost, widespread use, good foaming capabilities and low immunogenicity. Gelatine is a denatured protein synthesized from the partial hydrolysis of collagen from animal skin, bone, and tissue¹. It has the capacity to gel or remain a liquid due to its size and structure, which is important for making Clinisponge®/Tidespon absorbable hemostat. Type A gelatine is obtained from acid-treated collagen, and type B gelatine is obtained from an alkali-treated precursor².

Table 4: Components/Materials present in device with their body contact

Material	Composition/Grade	Purpose	Type of Body Contact	Supplier Information
Gelatine	Type A	A gelling agent that helps in hemostasis during use of the device.	Direct contact with tissue/bone/blood	P.B Gelatine (Manufacturer). Distributor: Kaizer Mackay
Formaldehyde	Formaldehyde solution of 37% stabilized by 10% methanol	Acts as a cross-linking agent during manufacturing process.	Direct contact with tissue/bone/blood	Merck İlaç Eczacı ve Kimya Tic.A.Ş. (Manufacturer)
Distilled Water	-	Acts as a vehicle to achieve a homogenous mixture of formaldehyde and gelatine	Direct contact with tissue/bone/blood	In-house

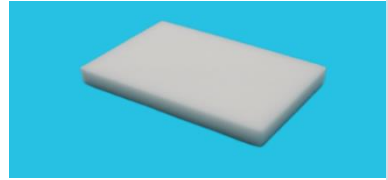
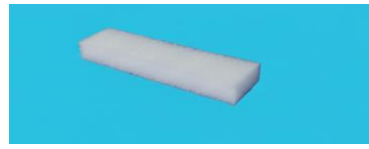
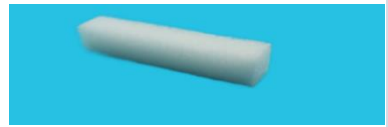
6.2 Physical characterisation

Gelatin-based absorbable hemostat is a medical device used for controlling bleeding during or after surgery. The material and its design promote blood clotting process and facilitates hemostasis at the site of application.

Table 5: Physical properties of different size CLINISPONGE®/TIDESPON

Sr. No	Model	Size	Description	Photo
1.	CLINISPONGE® /TIDESPON Standard	80 x 50 x 10 mm 70 x 50 x 10 mm	CLINISPONGE/TIDESPON Standard is usually used in all types of surgery; e.g. general surgery, thoracic surgery, in parenchymal surgery, etc. It can be cut into desired sizes.	
2.	CLINISPONGE® /TIDESPON Special	80 x 50 x 1 mm 70 x 50 x 1 mm 125 x 80 x 1 mm 250 x 80 x 1 mm	It is suitable for use where a thin layer of sponge is required for surgical procedures such as laminectomy, laparoscopy. It is also convenient to use in vascular surgery and plastic surgery fields.	

Sr. No	Model	Size	Description	Photo
3.	CLINISPONGE®/TIDESPON Film	200 x 70 x 0.5 mm 80 x 50 x 0.5 mm	To provide bleeding control in plastic surgery, vascular surgery, neurosurgery and on the large surfaces or on the places where it is difficult to reach.	
4.	CLINISPONGE®/TIDESPON Dial	30 x 30 x 10 mm 40 x 10 x 10 mm	It is convenient for the fistula applications of dialysis procedures to use.	
5.	CLINISPONGE®/TIDESPON Tampon	80 x Ø 30 mm 60 x Ø 20 mm	It is used mainly in anorectal surgery e.g. For anal fistulas and fissures by using a proctoscope. The tampon has a hole which a drain may be inserted.	

Sr. No	Model	Size	Description	Photo
6.	CLINISPONGE®/TIDESPON Dental	10 x 10 x 10 mm 14 x 7 x 7 mm	Used after tooth extraction, root amputations, removing of cyst, etc. in dental-oral surgery, mandibular-maxillar surgery.	
7	CLINISPONGE® Long/TIDESPON Long	125 x 80 x 10 mm	It is used in general surgery, abdominal surgery, orthopedic surgery, gynecological surgery, thoracic surgery.	
8	CLINISPONGE® Strip/TIDESPON Strip	80 x 20 x 10 mm	It is used in cases of after tooth extractions, for wound care of superficial wounds, and ossicular reconstruction.	
9	CLINISPONGE® Large/TIDESPON Large	250 x 80 x 10 mm	It is used in areas with extensive bleeding, such as abdominal surgery and orthopedic surgery.	
10	CLINISPONGE® Nasal/TIDESPON Nasal	50 x 10 x 10 mm	It is used in a variety of nasal surgeries, septoplasty, turbinectomy, rhinoplasty, after tooth	

Sr. No	Model	Size	Description	Photo
			extractions, and in superficial wounds.	
11	CLINISPONGE®/TIDESPON Powder	1gram	For the purpose of controlling the bleeding of irregular surfaces, such as in orthopedic surgery is used. It may be used dry or added sterile sodium chloride solution to make a paste for smearing on the surface.	

6.3 Chemical characterisation

The chemical information of the device shall be gathered through information provided by the supplier and a chemical characterisation test performed as per the standard ISO 10993-18:2020/Amd 1:2021 and ISO 10993-17:2023.

In this section mainly focus on qualitative and quantitative information of materials used in device. The supplier has provided the Material Safety Data Sheet (MSDS) and Certificate of Analysis (COA), which detail the chemical composition, material information, and associated safety data for the device.

6.4 Quantitative information

The quantitative chemical formulations of the remaining materials coming in either direct or indirect contact are further considered and evaluated for biological safety risk assessment.

Table 6: Composition of CLINISPONGE®/TIDESPON

Raw materials found in component	Proportion found in the finished component	C.A.S Registry Number
Gelatin	%100	9000-70-8
Formaldehyde	< 10ppm	50-00-0

6.5 Prior biological safety data review

CLINISPONGE®/TIDESPON manufactured by YÜCEL MEDİKAL VE TEKSTİL ÜRÜNLERİ SAN. TİC. LTD. ŞTİ. are similar to the following devices already in the market and approved:

Table 7: Similar devices of CLINISPONGE®/TIDESPON

Manufacturer	Similar Device Name	Regulatory Approval
Pfizer, Inc.	GELFOAM	CE0086
Ethicon, Inc.	SURGIFOAM	CE0543
Ethicon, Inc.	SPONGISTAN	CE0543

The review of prior biological safety data was conducted in accordance with ISO 10993-1 and ISO 14971 principles to identify potential biological risks associated with the Clinisponge®/Tidespon. The strategy for this review included:

- **Collection of Documentation:** Supplier-provided Material Safety Data Sheets (MSDS), Certificates of Analysis (COA), and technical datasheets for each material component (Gelatin, Formaldehyde and Distilled Water) were gathered and evaluated.
- **Cross-Referencing with Standards:** Material information was compared against applicable international standards (e.g., ISO, ASTM, USP) to confirm regulatory acceptance and historical use in medical devices.
- **Toxicological Profile Screening:** Available toxicological data from MSDS/COA were screened for hazards related to cytotoxicity, sensitisation, irritation, Material medicated toxicity, Acute systemic toxicity, sub-chronic and sub-acute systemic toxicity, Implantation, Genotoxicity, hemolytic effect and other relevant endpoints.
- **Historical Use Assessment:** Literature data and prior regulatory submissions were reviewed to assess clinical history and prior use of identical or similar materials in marketed devices.

- **Gap Analysis:** Potential data gaps were identified where supplier information was insufficient, and these gaps were addressed by integrating published literature, existing test data

6.5.1 Literature Search Strategy

The objective of this literature review is to assess whether sufficient published data are available to identify and evaluate potential biological risks associated with the materials of the Clinisponge®/Tidespon. The review specifically addresses the device components, including Gelatin, Formaldihyde and distilled water that may come into direct contact with tissue/bone/blood. The focus is on their biocompatibility, clinical applications, and potential toxicological concerns. The intent is to determine whether existing scientific evidence is adequate to support the identification and assessment of relevant biological endpoints in accordance with ISO 10993-1:2018 and related standards, thereby ensuring the relevance of the data to patient exposure during surgical procedures.

A systematic literature search will be conducted using scientific databases and search engines. The primary sources of literature include PubMed (peer-reviewed biomedical publications) and Google Scholar (broad scientific coverage). Additional relevant regulatory or scientific publications may be included where appropriate.

The databases for literature search include, but are not limited to, the following databases

Table 8: Data base and key words

Search engine	Data base	Key words
Google / Google scholar	PubMed	Review of gelatin, Formaldihyde used in Hemostate materials, Collagen and Gelatin, Absorbable Gelatin sponge in experimental surgery and applications
	PubChem	
	ECHA's Chemical Database	
	TOXNET	
	US FDA	
	EMA	

Documentation and inclusion criteria

- **Inclusion:** Peer-reviewed studies, systematic reviews, meta-analyses, regulatory guidance, post-market surveillance data, and relevant preclinical/clinical studies

addressing safety, toxicity, or biocompatibility. Included drugs and devices both the data base for literature search.

- **Exclusion:** Non-peer-reviewed sources, non-medical engineering papers without biological safety relevance, duplicate publications.

Outcome: The retrieved literature will be screened for relevance and quality. Selected publications will be critically evaluated and summarised to support a risk-based assessment of the Clinisponge®/Tidespon.

7. Manufacturing Process, Packaging, and Sterilisation

All the variants of CLINISPONGE®/TIDESPON in the scope of the technical documentation are manufactured by YÜCEL MEDİKAL VE TEKSTİL ÜRÜNLERİ SAN. TİC. LTD. ŞTİ.

The manufacturing flow chart for CLINISPONGE®/TIDESPON shown in figure 1:

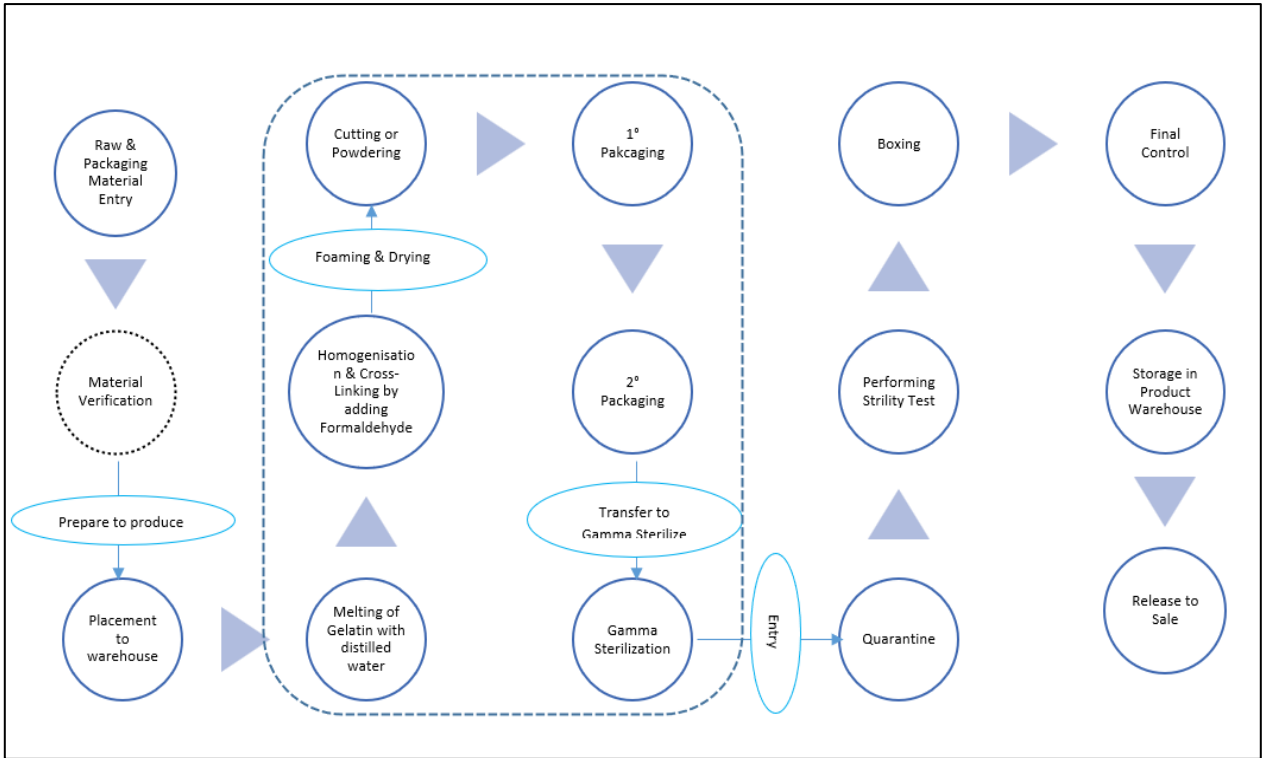


Figure 1: Manufacturing Flow Chart of CLINISPONGE®/TIDESPON

- **Manufacturing/processing:**

The manufacturing process of CLINISPONGE®/TIDESPON will be reviewed and any biological risk will be evaluated. The entire manufacturing process is carried out under controlled environment (clean room condition and in controlled area).

All the steps involved in the manufacturing process are performed for all raw and packaging materials, and in-process testing at various stages of manufacturing is carried out.

- **Cleaning process:**

No cleaning process is involved in the manufacture of CLINISPONGE®/TIDESPON.

- **Packaging material, including any printing ink:**

No printing is done on the device.

- **Sterilisation:**

The device is sterilised by the GAMMA sterilisation method.

The CLINISPONGE®/TIDESPON is a single-use, sterile device. It is packed into three levels of packaging: primary, secondary, and tertiary. The device is packed in a medical grade blister paper pouch, which indicates its primary packaging material comes into direct contact with the device.

Table 9: Packaging material

Packaging Type	Description	Material
Primary packaging	Blister-Tyvek pouch	Medical grade blister-paper
Secondary packaging	Folding carton box	Carton box
Tertiary packaging	corrugated box	Corrugated box

8. Hazard Analysis

Using available information, the biological risk is evaluated for each identified hazard and hazardous situation using a 3x3 matrix, which categorises the severity of risk and probability of occurrence.

In the absence of data on the probability of harm occurrence, risk will be evaluated based solely on the severity of the harm.

The risk evaluation was conducted using a 3x3 matrix, where Risk = Probability × Severity. Probability was categorised as High, Medium, or Low, and severity was categorised as Low, Medium, or High.

- **High Risk:** Likely to occur and/or could cause serious or life-threatening harm; requires immediate attention.
- **Medium Risk:** May occur and/or could cause moderate harm; should be monitored and controlled.
- **Low Risk:** Unlikely to occur and/or causes minor harm; acceptable with routine monitoring.

9. Endpoint analysis

ISO 10993-1:2018 provides guidance for the biological evaluation of medical devices within a risk management process. Annex A includes Table A.1, which outlines biological endpoints to consider based on body contact type and duration of exposure. This table serves as a framework for selecting relevant endpoints, not a mandatory checklist of tests. Manufacturers are expected to provide a scientific rationale when relying on existing data instead of conducting new biocompatibility tests.

Categorisation of a medical device according to ISO 10993-1

Medical devices shall be categorised according to the nature and duration of body contact as specified in the ISO 10993-1:2018. The categorisation of medical devices facilitates selection of appropriate data sets as described in the informative Annex A of this standard.

- **Nature of Body Contact:** Implant medical device (Direct contact with Tissue/Bone)
- **Contact Duration:** B (prolonged contact >24 to 30 d)

Table 10: Biological test identification table

Device name	Category	Biological endpoints	EN ISO 10993 Reference Guideline
CLINISPONGE /TIDESPON	Implant medical device coming into prolonged contact (>24 to 30 d) with tissue/bone/blood	Cytotoxicity	ISO 10993-5:2009
		Sensitisation	EN ISO 10993-10:2021
		Irritation test	EN ISO 10993-23:2021
		Material mediated pyrogenicity	EN ISO 10993-11:2017
		Acute systemic toxicity	EN ISO 10993-11:2017
		Sub-acute toxicity	EN ISO 10993-11:2017
		Implantation effects	ISO 10993-6:2016
		Hemocompatibility	ISO 10993-4:2017/Amd 1:2025
		Genotoxicity	OECD 471 OECD 487 OECD TG 471

10. Testing Rationale

Based on the literature review and hazard analysis, the following tests are planned to be conducted on the Clinisponge/Tidespon to evaluate its biological safety in accordance with ISO 10993 series.

11. Chemical Characterisation as per ISO 10993-18

In accordance with ISO 10993-18, chemical characterisation will be performed. If any new or unexpected substances are identified, a toxicological risk assessment will be done.

12. Toxicological Risk Assessment Data

Based on the results of the chemical characterisation study, a Toxicological Risk Assessment (TRA) will be conducted for all identified substances to evaluate potential biological risks and ensure the safety of Clinisponge/Tidespon.

13. Conclusion

The Biological Evaluation Plan (BEP) provides reasonable confidence that the existing data, together with the results from the proposed tests, will demonstrate the biological safety of the device for its intended clinical use.

14. References for Literature review

1. Irfan, N. I., Mohd Zubir, A. Z., Suwandi, A., Haris, M. S., Jaswir, I., & Lestari, W. (2022). Gelatin-based hemostatic agents for medical and dental application at a glance: A narrative literature review. The Saudi Dental Journal, 34(8), 699–707. <https://doi.org/10.1016/j.sdentj.2022.11.007>
2. Hanani, Z. A. N. (2015). Gelatin. In Encyclopedia of Food and Health (pp. 191–195). Elsevier Inc. <https://doi.org/10.1016/B978-0-12-384947-2.00347-0>

Annexes

Appendix 1: Material safety datasheet

Appendix 2: Literature study

Appendix 3: Biocompatibility tests reports with chemical characterisation

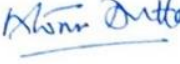
Appendix 4: Test reports (BET, Gamma sterilisation, Clean room validation)

Appendix 5: CV's information of Author and reviewer

Revision history

Version No.	Date	Description
00	07/03/2023	Initial release: Transition from MDD 93/42/EEC to EU Medical Device Regulation 2017/745
01	02/12/2025	Added additional variant of CLINISPONGE®/Tidespon. Added Basic UDI DI of an additional variant of CLINISPONGE®/Tidespon Added ® symbol for every clinisponse variant. Template modified with removed procedure of biological safety evaluation and added literature search strategy.

Document approvals

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Name	Bhargavi Raj	Atonu Dutta	Şenol YÜCEL
Signature and Date	 01/12/2025	 02/12/2025	 02/12/2025
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